

Chapter 1

Weeks 1

ECEN-689 Dr. Zhiwen Fan TTh 9:35 AM - 10:50 AM AIEN M113

1.1 Introduction – 1/13

Geometry, semantics, affordance, actionable information

1.2 Introduction & Guest Lectuer - 1/15

Hezhen Hu performance drops for users and contexts in the long tail inclusion: how to build models that generalize to diverse humans across bodies, context core: human foundational models embodiment -> inclusive; understanding -> context-aware general visual benchmark million-level (datacomp, panda) human-centric datasets: thousand-level (CSL-daily-21k; dymvhuman) training human foundation model (web data - vast scale, low quality) BERT success in MLP; can this pretraining strategy work for human video

words: rich semantics, discrete, high-level video: low info densit, continuous, redundant

three key components: visual token; pretraing strat; human prior visual token: compact human pose (input -> visual token -> GCN -> input embedding) filters irrelevant information reduces redundancy (reduce input dimension by 2,000+ times) multi-level masking (randomly mask joints; randomly mask frames; randomly mask frames) prelim: MANO: parametric hand model MANO is pre-learned from large hand scans (provide mapping) MPL -> MANO ()& proj) -> projection -> hand reconstruction SignBERT shows versatile capability on multiple sign language understanding tasks

EVA: expressive human modeling from in-the-wild video #1 expressive animatable avatar #2 simple rgb video input (no human mesh annotation required)

monocular rgb video -> smpl-x alignment -> gaussian avatar modeling -> animate avatar multi esimators: 2d pose (dwpose) & 3d body (smpl-x) & 3d hand (hamer) all goes to iterative fitting to extract each models complimentary strengths

HumanNOVA: efficient avatar modeling from a single image data: scalable data generation pipeline model: scalable transformer backbone

data: synthetic data generation (rigged human mesh -> (smpl-x pose) -> animated mesh & render) real-world data generation (multi-view captured data -> (3dgs fitting) -> 3dgs assets &

render)

model: multi-modal encoding; incorporate human coarse mesh as human prior 2d-to-3d mapping; triplane as 3d representation; updated from multi-modal encoding via cross-attn

MMHU: scalable data annotation pipeline * large-scale video collection (paid/collected; YouTube video; Waymo video) * into low-level context (pose & description) goes into * high-level semantics (behavior & intent)

Future: * human-centric 3d foundation model: current foundation model only sees people; desired model understands people in 3d and context 1. scalable data infrastructure -> human-centric 3d foundation model -> diverse downstream tasks 2. human-centric 3d foundation model (unified model): structured human encoder -> unified latent space (geo, contact, motion, context, intent) 3. 3d human understanding & gen, interaction & intent prediction, ...

* embodied active reasoning (from passive perception to active human understanding) passive vision model (static camera) leads to unsafe action active agent (dynamic interactives) leads to SAFE action observe, reason: belief-aware human state representation -> active reasoning

* embodied social affordance learning: enable agents to support effective and inclusive multi-person collaboration agent needs to understand role of each person and their reactions read social cues in (understanding) -> reason & react with social knowledge (reason) -> output predictions in more socially-suitable way!

main lecture:

current state of the art: many less than 5 years * cv still active research area, rapidly changing * deep learning and generative methods powering many modern applications

why is cv difficult -> viewpoint variation, illumination, scale * also intra-class vision (concepts), motion, background clutter, occlusion * perception is an inherently ambiguous problem; must often use prior knowledge about the world's structure

1.3 Week 2 – Jan 22

1.4 Week 2 – Jan 22

1.4.1 Student 1

BEVFormer

In autonomous driving, car needs to understand the whole scene around it, using cameras that each see only part of the world.

Need a unified, ego-centric representation that puts everything into one shared view.

BEV Queries, grid where we want to know what is present.

BEV architecture: two key architectures: spatial cross-attention and temporal self-attention.

Sparse cross attention: full BEV-to-image cross-attention is too expensive. BEVFormer proposes sparse spatial cross-attention. Each BEV query efficient uses most relevant camera.

Temporal self-attention: fuses history in a recurrent way, since pure memory would be too costly

BEVFormer builds a unified BEV representation from multi-camera images with a query-based transformer. Directly constructs BEV.

VoxFormer

Model needs to predict voxel grid in 3D space. If object is identified, needs to predict semantic class. Model infers in visible surface and occluded space as well so 3D volume is complete, model has to reason occluded geometry.

Compared to BEV-based detection. Natively apply BEV feature, still much heavier, use sparse voxel representation. Use important elements of query proposals to reduce computational cost. Result is improved performance over MonoScene since only relevant query proposals is sent to transformer, helps model focus on important 3D locations.

2D features are extracted by ResNet50 from RGB images, and depth is estimated by depth predictor.

VoxFormer targets camera-based 3D semantics scene completion and makes 3D reasoning feasible via sparse voxle tokens. Key design: depth/occupancy-guided query proposal + deformable attention, then mask-token completion in voxel space.

1.4.2 Student 2 – Dayou Li – Latent Action

Bottleneck in robot learning: large-scale robot datasets are expensive to collect, require robot hardware, but contain robot action. Internet-scale video data is widely available, but there is a human to robot gap, and no included robot information.

LAPA – Latent action pretraining from video

Latent action quantization: change in current to observation to predict latent action (VLM) LAPA determines action. 1. latent action quantization. 2. latent pretraining 3. LAPA $\rightarrow$ action.

Latent action encoding: input of current and next frame, patch embed, spatial transformer, causal transformer which outputs transition of two embeddings.

$$z_1 = \arg \min_{z_k} |d_1 - z_k|^2$$

$$L = ||x_2 - \hat{x}_2||^2$$

This model is compared to OpenVLA, ActionVLA, outperformed them.

Reconstruction loss is supervision. Want latent actions to predict.

CLAP model

Supervised in both reconstruction in visualization and action space.

Pipeline: Act-VAE action trajectory $\rightarrow$ codebook $\rightarrow$ decode action

VD-VAE: visual feature of the current frame and another frame, encoded by Dyno (can use another pretraining encoder), outputs two branches $z_{q,a}$ and $z_{q,i}$, these two branches go to encoder with first frame and outputs last frame as the supervision.

CLAP-NTP: VLM to turn obs and instr to subtask and action tokens

CLAP-RF: uses VLM backbones that encode current vision and language context

Conclusions: latent actions provide a scalable token interface for learning from videos, which is valuable for cross-embodiment learning. LAPA shows that video-only tokenization can work and can leverage human videos, but tokens may be visually entangled and require action finetuning for executability. CLAP strengthens grounding by aligning video latents to robot actions latents and adds a planner-controller style.

1.5 Week 3 – Jan 27

Text-to-image generation; DALL-E 2;

Textual token: turn input text into tokens, encoder needs to be fully well-trained with web-scale data. CLIP objective is the go between to the image encoder.

DALL-E 3;

GPT-4;

Segment Anything Model (SAM), large image model, very high-scale segmentation.

1.5.1 Neural networks – the original linear classifier

(Before) Linear score function: $f = Wx$. (Now): 2-layer neural network: $f = W_2 \max(0, W_1 x)$. where $x \in \mathbb{R}^D$, $W_1 \in \mathbb{R}^{H \times D}$, $W_2 \in \mathbb{R}^{C \times H}$.

ReLU still used commonly as activation function, good default choice for most problems. Leaky ReLU has small weight on left side.

Forward processing is done at inference time. Backward processing is done at training time.

Computing gradients

$s = f(x; W_1, W_2) = W_2 \max(0, W_1 x)$ nonlinear score function;

$L_i = \sum_{j \neq y_i} \max(0, s_j - s_{y_i} + 1)$ hinge loss on predictions;

$R(W) = \sum_k W_k^2$ regularization;

$L = \frac{1}{N} \sum_i L_i + \lambda R(W)$ total loss: data loss + regularization

Backpropagation in Neutral Networks

$$f(x, y, z) = (x + y)z$$

$$q = x + y$$

$$f = qz$$

Then, by chain rule,

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial q} \frac{\partial q}{\partial x} = z \cdot 1 = z$$

$$\frac{\partial f}{\partial y} = \frac{\partial f}{\partial q} \frac{\partial q}{\partial y} = z \cdot 1 = z$$

$$\frac{\partial f}{\partial z} = \frac{\partial f}{\partial q} \frac{\partial q}{\partial z} = q \cdot 1 = x + y$$

Regularization intuition: prefer simpler models

When the model starts to overfit, training accuracy rises, but validation accuracy drops.

Regularization:

$$L(W) = \frac{1}{N} \sum_i \underbrace{L_i(f(x_i, W), y_i)}_{\text{dataloss : model prediction should match training data}} + \underbrace{\lambda R(W)}_{\text{regularization: prevent the model from overfitting}}$$

Components of CNNs

convolutional layers: normalization layers $\hat{x}_{i,j} = \frac{x_{i,j} - \mu_j}{\sqrt{\sigma_j^2 + \epsilon}}$;

pooling layers; dropout (sometimes); fully-connected layers; activation functions (ReLU, sigmoid, tanh, softmax).

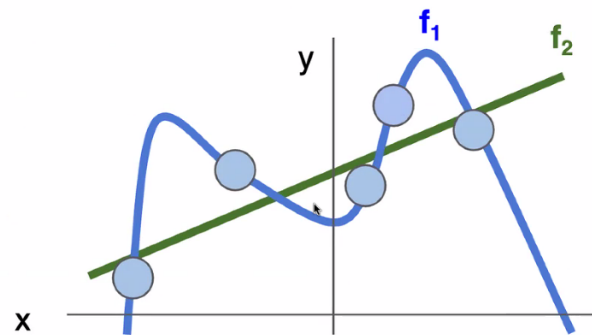


Figure 1.1: Regularization prefers simpler models that generalize better.

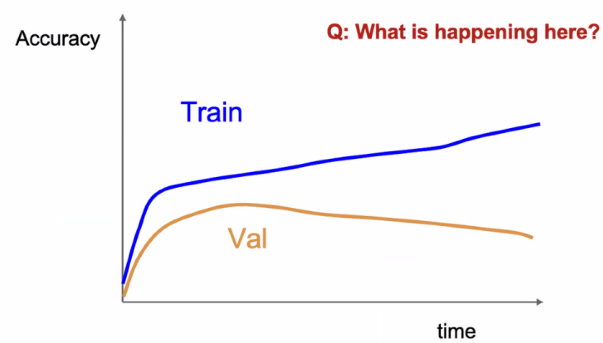


Figure 1.2: Regularization prefers simpler models that generalize better.

Chapter 2

Full 3D Perception and Reconstruction Pipeline

Input images -> feature extraction -> feature matching -> estimate camera poses -> ...

2.1 Capturing Reality – in 3D Point Cloud

3D panning from just plain input images. Sparsely sampled input data can produce smooth views of the same scene. Use images to build neural 3D image representation.

NeRF can render highly realistic images: input images $\rightarrow$ optimization $\rightarrow$ rendering.

2.2 Three key components

Neural volumetric 3D scene representation, differentiable volumetric rendering function, optimization via analysis-by-synthesis.

MLP at initial layer to store information about the environment

$$(x, y, z, \theta, \phi) \rightarrow F_{\Omega} \rightarrow (r, g, b, \sigma)$$

x, y, z are the spatial direction and θ, ϕ the viewing direction. F_{Ω} has 9 layers, 256 channels.

It's not a mesh, not a point cloud either it's continuous voxels made of shiny transparent cubes.

Can use an MLP to represent your image. By using a small MLP, you can effectively compress your initial image. MLP results in a loss of sharpness as small MLPs cannot remember every detail, resulting in blur.

2.2.1 Cameras and rays

We need the mathematical mapping from (camera, pixel) $\rightarrow$ ray. Then we can abstract underlying problem as learning the function ray $\rightarrow$ color (the "plenoctic function").

Coordinate frames + transforms: world-to-camera

Extrinsics (R, T) (rotation, translation) brings world coordinates to camera coordinates. Intrinsic (K) bring camera coordinates to image coordinates. The reverse process is also possible for both of these steps.

Neural volumetric rendering

Continuous, differentiable meaning.

Surface vs. volume rendering

We want to know how rays interact with scene. Surface rendering – loop over geometry, check for ray intersection. Volume rendering – loop over ray points, query geometry.

Volume rendering: rendering model for ray $r(t) = o + td$:

$$C \approx \sum_{i=1}^N T_i \alpha_i c_i$$

How much light is blocked along ray:

$$T_i = \prod_{j=1}^{i-1} (1 - \alpha_j)$$

How much light is contributed by ray segment

$$\alpha_i = 1 - e^{-\sigma_i \delta t_i}$$

2.3 Rendering: projecting 3D to 2D

Neural radiance field is a subset of neural rendering. Typically have 3 stages. Let the pixels find what features they need.

2.3.1 Neural radiance field (NeRF)

$$\arg \min_V \sum_i \left\| \mathcal{R}(V; R^{(i)}, t^{(i)}, P^{(i)}) - y \right\|_2^2$$

NeRF is too slow for real-time rendering. Instead, for every pixel, query all 3D points from MLP. Rendering 1 pixel, you'll have 10 or 100 queries. It is nearly impossible to have real-time even with powerful GPUs. Original NeRF rendering speed is 0.06 FPS, has been improved up to 15.0 FPS (DNeRF).

Chapter 3

3D Gaussian splatting

Able to dratically improve render time. Up to 135.0 FPS from NeRF's 0.071 FPS.

3.1 Challenges with 3D-GS

Significant amount of disk spaces are required in the explicit representation. Excessive Gaussian number, often millions on each scene. Next idea: for every 3D point, we are not using just RGB, we are using spherical harmonics. These coefficients requires high dimensions. Last, using `float32` is very heavy.

3.1.1 Excessive Gaussian number

Many 3D points have small opacity. If its opacity is very small, its contribution is small, so it can be ignored. So preserve those points with high opacity, and remove those with low opacity to improve performance.

$$GS_j = \sum_{i=1}^{MHW} 1(G(\mathbf{X}_j), r_i) \cdot \sigma_j \cdot \gamma_j(\Sigma),$$
$$\gamma(\Sigma) = (V_{\text{norm}})^\beta$$
$$V_{\text{norm}} = \min \left(\max \left(\frac{V(\Sigma)}{V_{\text{max90}}}, 1 \right), 1 \right)$$

Chapter 4

February 17 Lecture

4.1 3D scene understanding

For homework: 20 dB (higher is better), above 0.8 SSM; downsample image by 2.

Semantic segmentation typically takes place in 2D. Fully convolutional networks for segmentation. UNet paper.

4.1.1 UNet

Originally we have a high resolution image, we can downsample to lower and lower size image. Downsampling and upsampling for pyramid perception. Stacking many many layers, gradient becomes problematic. ResNet proposes solution that resolves this issue.

4.1.2 3D shape completion: ScanComplete

Fill in missing data from initial scan since some regions will not be in the scan (under table).

Larger scale 3D datasets. ScanNet++:

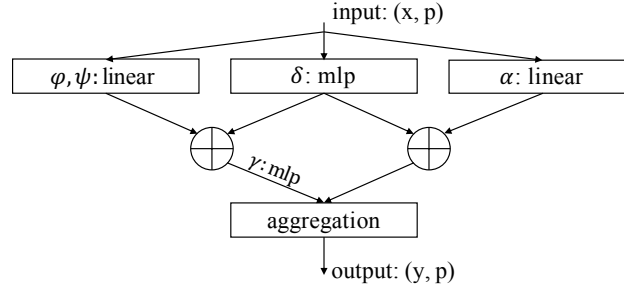
1. 1006 high-fidelity scenes
2. 1mm laser scans - DSLR images
3. iPhone RGB-D-semantics

4.1.3 3D point cloud segmentation: PointNet

End-to-end learning for scattered, unordered point data. Unified framework for various tasks. Classification network: $n \times 3 \rightarrow n \times 64 \rightarrow n \times 1024 \rightarrow k$ or $n \times k$ output scores. MLP based neural network.

4.1.4 Point transformer: the basic unit

Using transformer to process 3D point cloud. Architecture is similar to UNet: downsample first, then upsample.



4.1.5 NeRF with semantic segmentation

Input is $\text{PE}(x, y, z)$ output is semantic (s) and RGB (c).

$$\hat{\mathbf{C}}(\mathbf{r}) = \sum_{k=1}^K \hat{T}(t_k) \alpha(\sigma(t_k) \delta_k) \mathbf{c}(t_k),$$

where

$$\hat{T}(t_k) = \exp \left(- \sum_{k'=1}^{k-1} \sigma(t_k) \delta_k \right)$$

And

$$\hat{\mathbf{S}}(\mathbf{r}) = \sum_{k=1}^K \hat{T}(t_k) \alpha(\sigma(t_k) \delta_k) \mathbf{s}(t_k)$$

where

$$\hat{T}(t_k) = \exp \left(- \sum_{k'=1}^{k-1} \sigma(t_k) \delta_k \right)$$

4.1.6 3D data are rare, image data are everywhere

Foundation models thrive on large, diverse datasets. Annotated video is rare, but image data is publicly available virtually everywhere.

4.1.7 Can we achieve 3D segmentation without labels?

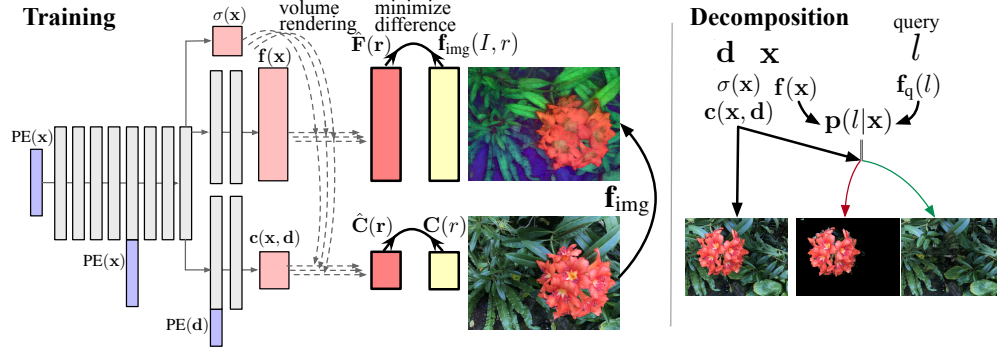
Reference paper: Decomposing ReRF for Editing via Feature Field Distillation.

For each 3D point we have a feature, then we do the volume rendering.

4.1.8 NeRF is slow, but we can accelerate this with 3DGS

Reference paper: Feature 3DGS: Supercharging 3D Gaussian Splatting to Enable Distilled Feature Fields.

In this work, we supercharge the 3D gaussian splatting to learn feature fields via knowledge distillation from 2D foundation models.



4.1.9 Unified 3D reconstruction with Gaussian Splatting

Reference paper: Large Spatial Model: End-to-end Unposed Images to Semantic 3D.

We have some fixed views, but we can output images from a changed perspective*.

4.1.10 Text-Driven Semantic Queries over 3D Maps

Reference paper: Large Spatial Model: End-to-end Unposed Images to Semantic 3D.

Collect many images from different scenes, train model on these many images to be able to generalize. Input is just image, no annotation or other information. Model has to find camera pose and how to map pixel to 3D gaussian. Find black box that maps pixel to 3D gaussian. Then it can warp from one image to another to find the relative information based on the data.

4.1.11 Cross-Modal Fusion for Semantic 3D Gaussians

Reference paper: Large Spatial Model: End-to-end Unposed Images to Semantic 3D.

Also render the feature map alongside the 2D image. This is using a transformer model.

Chapter 5

The 3D World

Feb. 24

Dynamic subjects, static 3D environment, and the observer (camera pose) are information we need to recover.

5.1 Mapping Applications

Indoor mapping, undersea mapping, underground mapping, and space mapping. The robot needs to build a perception system and make decisions with limited computing power. The robot needs a concept of its surrounding, that is why 3D mapping is so important.

5.2 Simultaneous Localization and Mapping (SLAM)

Given the robot's controls and observations, get a map of the environment and the path of the robot. Building a map and locating the robot in the map at the same time (chicken-and-egg problem).

5.2.1 Why SLAM is Hard

Robot path error correlates errors in the map.

In the real world, the mapping between observations and landmarks is unknown. ... wrong data associations can have catastrophic consequences

5.2.2 SLAM Categories

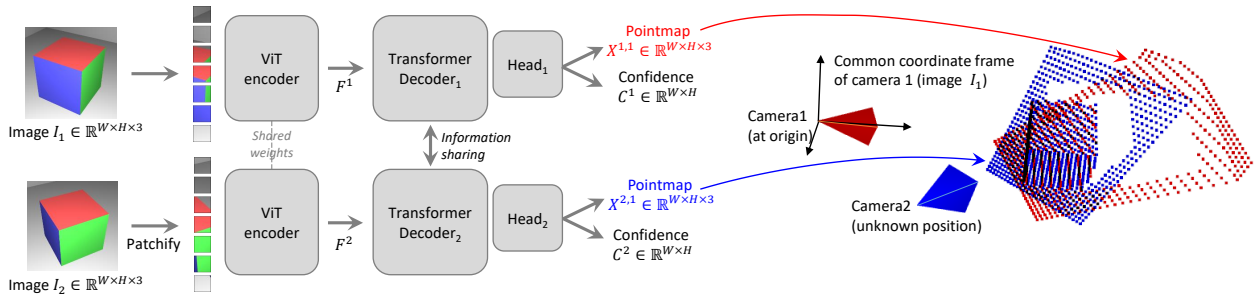
SLAM Maps can be categorized as either sparse or dense. Achieving real-time dense construction with just single camera is challenging. Existing real-time monocular SLAM systems use mostly sparse methods.

Sparse SLAM method, such as ORB-SLAM track a handful of discrete, distinctive keypoints. These methods are lightweight, fast, and provides an accurate camera pose tracking in real time. However, the resulting maps are under modelled (less-detailed).

On the other hand, dense SLAM approaches estimate per-pixel level accurate features from depth across the entire image. This creates a detailed representation of the environment and predicts accurate camera trajectory. However, it is computationally intensive which limits them to operate real time compared to sparse methods.

5.3 Front End

Closed-loop training using 3D Gaussian and the reference image. 3D Gaussians are used in Dense SLAM, as the entire scene is captured, rather than in Sparse SLAM, where only key points are captured.



DUST3R: learn a function $F : (I_1, I_2) \rightarrow (P_1, P_2)$. Mapping images $I_i[y, x] = (R, G, B)$ to point maps $P_i[y, x] = (X, Y, Z)$.

DUST3R made 2-frame prediction feasible, VGGT made 3-frame prediction feasible.

5.4 Video to 3D Reconstruction

Video $\rightarrow$ split to chunks $\rightarrow$ 3D reconstruction over chunks $\rightarrow$ alignment.

5.5 Excessive Gaussian Number

$$GS_j = \sum_{i=1}^{MHW} 1(\mathbf{G}(\mathbf{X}_j), r_i) \cdot \sigma_j \cdot \gamma_j(\Sigma)$$

where $\gamma(\Sigma) = (V_{\text{norm}})^\beta$ and $V_{\text{norm}} = \min\left(\max\left(\frac{V(\Sigma)}{V_{\text{max90}}}, 0\right), 1\right)$

5.6 Vectree Quantization

Paper: LightGaussian